

Program 8: Create Kubernetes Pods using YAML Descriptors

Project Structure

program8

- |--nodejs-configmap.yaml
- |--nodejs-pod.yaml
- |--[server.js](#)

Create a folder(p8)

```
Dec 12 10:26 AM
isha@meow: ~/DevOps/program8

isha@meow:~$ cd DevOps/
isha@meow:~/DevOps$ mkdir program8
isha@meow:~/DevOps$ cd program8
isha@meow:~/DevOps/program8$ nano server.js
isha@meow:~/DevOps/program8$ nano nodejs-app-config.yaml
isha@meow:~/DevOps/program8$ nano nodejs-pod.yaml

isha@meow:~/DevOps/program8$ minikube start
🐾 minikube v1.37.0 on Ubuntu 24.04
🌟 Using the docker driver based on existing profile
👍 Starting "minikube" primary control-plane node in "minikube" cluster
🔧 Pulling base image v0.0.48 ...
🔄 Restarting existing docker container for "minikube" ...
🔧 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔍 Verifying Kubernetes components...
   ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌞 Enabled addons: storage-provisioner, default-storageclass
🏡 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
```

Step 1: Write a [Node.js](#) app

nano [server.js](#)

```
GNU nano 7.2 server.js
const http = require("http");

const port = 3000;

const server = http.createServer((req, res) => {
  res.end("Hello from Node.js running inside Kubernetes!");
});

server.listen(port, () => {
  console.log(`Server running at port ${port}`);
});
```

Step 2: Create a ConfigMap for the code

```
GNU nano 7.2 nodejs-configmap.yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: nodejs-app-config
data:
  server.js: |
    const http = require("http");
    const port = 3000;

    const server = http.createServer((req, res) => {
      res.end("Hello from Node.js running inside Kubernetes!");
    });

    server.listen(port, () => {
      console.log(`Server running at port ${port}`);
    });
```

Step 3: Create the port yaml using [Node.js](#) as a base image

```
GNU nano 7.2 nodejs-pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: nodejs-app
  labels:
    app: nodejs
spec:
  containers:
    - name: nodejs-container
      image: node:18-alpine
      command: ["node", "/usr/src/app/server.js"]
      volumeMounts:
        - name: app-code
          mountPath: /usr/src/app/server.js
          subPath: server.js
      ports:
        - containerPort: 3000
  volumes:
    - name: app-code
      configMap:
        name: nodejs-app-config
```

Step 4: Apply the YAML files

Step 5: Check pod status

```
isha@meow:~/DevOps/program8$ kubectl apply -f nodejs-app-config.yaml
configmap/nodejs-app-config created
isha@meow:~/DevOps/program8$ kubectl apply -f nodejs-pod.yaml
pod/nodejs-pod created
isha@meow:~/DevOps/program8$ kubectl get pods
NAME                                READY   STATUS              RESTARTS   AGE
kubia                               1/1     Running            2 (87s ago) 3d23h
nginx-deployment-6f9664446b-n5tkx  1/1     Running            3 (14h ago) 10d
nginx-deployment-6f9664446b-nzg2z  1/1     Running            3 (14h ago) 10d
nodejs-pod                          0/1     ContainerCreating  0           13s
isha@meow:~/DevOps/program8$ kubectl port-forward nodejs-pod 3000:3000
error: unable to forward port because pod is not running. Current status=Pending
```

Step 6: Access the [Node.js](#) app

```
isha@meow:~/DevOps/program8$ kubectl get pods
NAME                                READY   STATUS              RESTARTS   AGE
kubia                               1/1     Running             2 (112s ago) 3d23h
nginx-deployment-6f9664446b-n5tkx  1/1     Running             3 (14h ago)  10d
nginx-deployment-6f9664446b-nzg2z  1/1     Running             3 (14h ago)  10d
nodejs-pod                          0/1     ContainerCreating   0           38s
isha@meow:~/DevOps/program8$ kubectl logs nodejs-pod
Error from server (BadRequest): container "program-8" in pod "nodejs-pod" is waiting to start: ContainerCreating
isha@meow:~/DevOps/program8$ kubectl get pods
NAME                                READY   STATUS              RESTARTS   AGE
kubia                               1/1     Running             2 (3m24s ago) 3d23h
nginx-deployment-6f9664446b-n5tkx  1/1     Running             3 (14h ago)  10d
nginx-deployment-6f9664446b-nzg2z  1/1     Running             3 (14h ago)  10d
nodejs-pod                          1/1     Running             0           2m10s
isha@meow:~/DevOps/program8$ kubectl port-forward nodejs-pod 3000:3000
Forwarding from 127.0.0.1:3000 -> 3000
Forwarding from [::1]:3000 -> 3000
Handling connection for 3000
```

Step 7: Check the output in <http://localhost:3000>



